Harnam Singh

Software Developer | JavaScript • TypeScript • Web Development
Kolkata, India | +91 80172 19644 | harnamsinghsandhu04@gmail.com | LinkedIn | GitHub | Portfolio

SKILLS

Languages: JavaScript, TypeScript

Frameworks / Libraries: Next.js, React, Tailwind CSS

Web APIs / Security: Web Crypto API, Client-Side Encryption

Tools / Platforms: Git, GitHub, localStorage

Concepts: Authentication, Secure Data Storage, Data Structures & Algorithms, Responsive Web Development

PROJECTS

iOS-Style Calculator Web App (Next.js, Tailwind CSS) – Live | GitHub

- Built responsive web calculator supporting 10+ arithmetic and expression operations including chained calculations, percentage, and sign toggle.
- Implemented custom two-stack expression evaluator enabling accurate operator-precedence handling and O(n) multi-step expression parsing.
- Designed mobile-first UI ensuring consistent usability across multiple screen sizes and devices. sizes.

Personal Secret Manager (Next.js, TypeScript, Web Crypto API) – Live | GitHub

- Built secure multi-user credential management application implementing client-side AES-256 encryption and PBKDF2-based authentication, ensuring sensitive data remains encrypted at rest.
- Developed modular cryptographic and storage layers enabling secure credential isolation per user and preventing plaintext exposure in localStorage.
- Structured application logic to allow easy future integration with backend databases without changing core encryption functionality.

CERTIFICATIONS & ACHIEVEMENTS

NPTEL – Computer Graphics (IIT Guwahati) | View Certificate

• Secured 80% (Top 1%)

JavaScript Algorithms and Data Structures – freeCodeCamp (Sep 2023) | View Certificate

EDUCATION

Master of Computer Applications (MCA), ***IGNOU***

JAN 2026 – DEC 2027 (ODL)

B.Sc. Mathematics (Honours), ***University of Calcutta***

2022 – 2025 | CGPA: 6.18 / 10